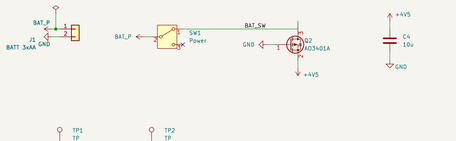
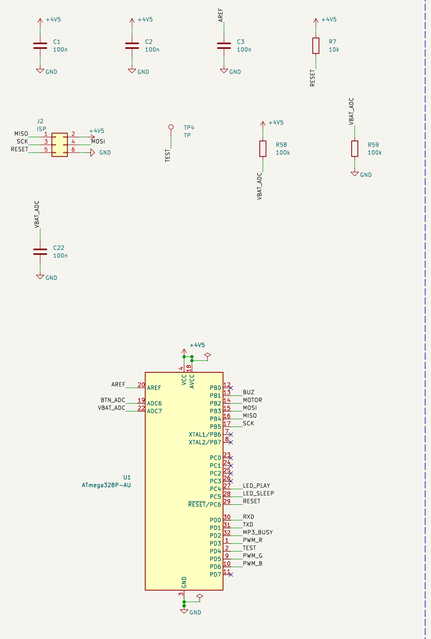
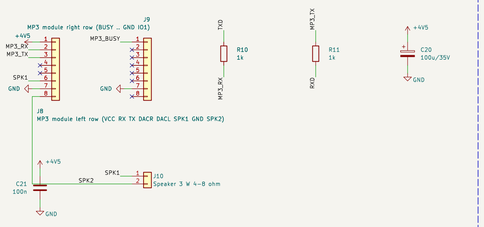
POWER: 3xAA, switch, reverse-polarity PMOS



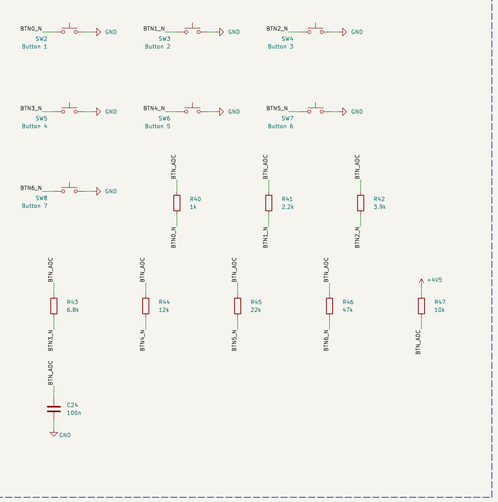
MCU: ATmega328P-AU, 8 MHz internal



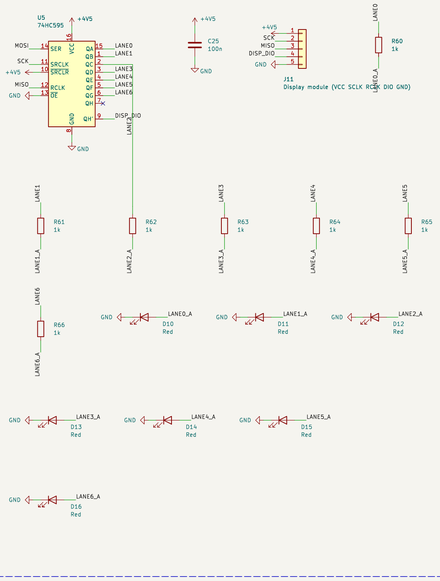
MP3 MODULE (DFPlayer Mini type) + SPEAKER



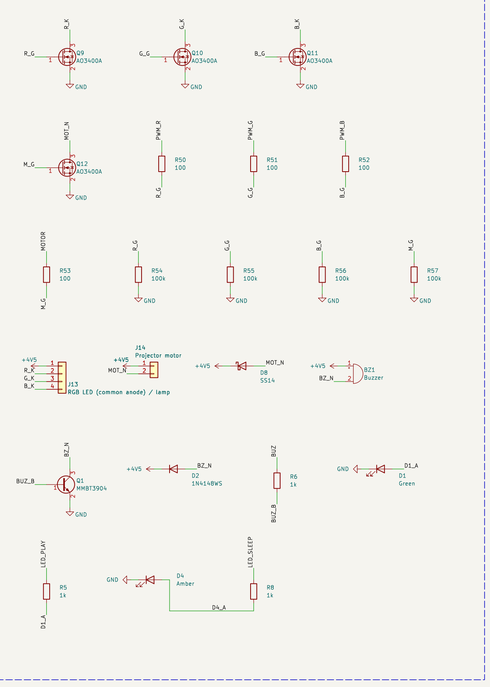
SEVEN BUTTONS ON ONE ADC (resistor ladder)



BALL LANE 74HC595 + 7 LEDs, DISPLAY MODULE LEAD



LAMP + MOTOR DRIVERS, BUZZER, LEDS



AP1 shared audio platform (rhymes player, star lamp, cricket game). ATmega328P at 8 MHz internal from 3xAA. MP3 module: 1 k series on both UART lines, BUSY on D2. Seven buttons share ADC6 through a ladder (10 k pull-up; 1 k … 47 k to each button); decode one press at a time in the HAL. Display: bought-in 4-digit 74HC595 LED module on J11 (VCC SCLK RCLK DIO GND). U5 74HC595 drives the seven ball-lane LEDs and its QH' feeds the module DIO, so one 24-bit frame is shifted per refresh: module segment byte, module digit byte, then the lane byte; latch on RCLK. SER/SRCLK/RCLK share MOSI/SCK/MISO with the ISP header (the module and U5 only see clocks while RCLK is low, harmless during ISP). Lamp: three AO3400A on PWM D3/D5/D6 to a common-anode LED; motor on D10 with SS14 flyback. Arduino tone() uses Timer2, which also drives D3: do not use the buzzer and red PWM together. Populate per product (see the manual).

First engineering release; see pcb/README.md for status and limits
ClaudeWork
Sheet: /
File: ap1_audio.kicad_sch

Title: AP1 shared audio platform (rhymes player, star lamp, cricket game)

Size: A1	Date: 2026-11-01	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1